# Tides

Most seashores on Earth experience rising and falling sea levels, known as tides, caused by the gravity of the orbiting Moon. The difference between high and low tide level varies from day to day, and from place to place. As the tides rise and fall, they also make the water flow along coasts in strong currents, first one way and then the other.

## SPRING TIDES

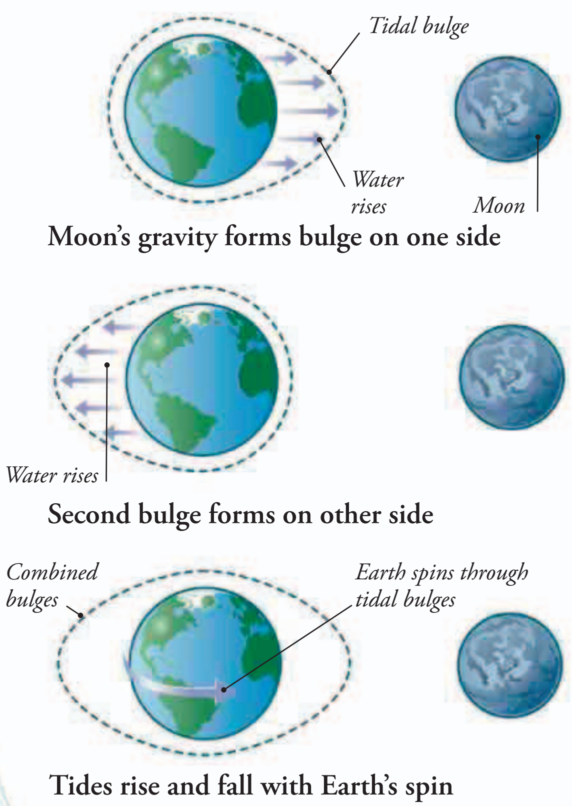
The Moon's gravity is the main cause of the tides, but the Sun's gravity also has an effect. When the Sun and Moon are in line, their gravitational pulls combine to create an extra-big tidal rise and fall. This is called a spring tide, and it happens twice a month—on the full Moon and the new Moon. During the twice-monthly half Moons, the gravity of the Sun pulls in a different direction, causing the much smaller neap tide.

## TIDAL STREAMS

As the tide rises and falls, it shifts water along coasts in tidal streams. On the rising tide the streams flow one way, and on the falling tide they flow the other way. This makes water travel difficult, because a boat crossing a tidal stream may be carried sideways as fast as it moves forward. It has to be steered in a different direction to compensate for this.

## LUNAR INFLUENCE

As the Moon orbits Earth, its gravity pulls water toward it in a tidal bulge. But Earth is also orbiting the Moon very slightly, and this slings water towards the other side of the world to create a second tidal bulge. As Earth spins, most shores pass in and out of both bulges each day, causing two high tides and two low tides.




▼ USING THE TIDE
*Tidal streams can flow as fast as a boat like this can sail, so it's vital to make the trip when they are flowing in the right direction.*